

ANSWER KEY — Integration Lesson 3 Test: Slicing Curves into Rectangles

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Rates at $t = 0, 1, 2, 3, 4$ min: 3, 5, 7, 9, 11 L/min. LEFT sum for $t = 0$ to 4, width 1?

Work: a left sum uses each slice's STARTING rate: 3, 5, 7, 9 (the $t = 4$ reading is never a left edge). Left sum $= 1 \times (3 + 5 + 7 + 9) = 24$. **Answer: about 24 liters.** (Classic error: adding all five readings $\rightarrow 35$, which uses too many rectangles.)

2. Same table: RIGHT sum, and which estimate is the overestimate?

Work: right sum uses each slice's ENDING rate: 5, 7, 9, 11 $\rightarrow 1 \times (5 + 7 + 9 + 11) = 32$. The rates are increasing, so right edges are the tall ones $\rightarrow$ the right sum is the overestimate (truth is between 24 and 32). **Answer: 32 liters; the right sum overestimates.**

3. LEFT sum for $y = 2x$ from $x = 0$ to 3, three slices of width 1.

Work: heights at left edges $x = 0, 1, 2$ are 0, 2, 4. Left sum $= 1 \times (0 + 2 + 4) = 6$. **Answer: 6.**

4. Same setup: RIGHT sum, true triangle area, and the sandwich check.

Work: right-edge heights at $x = 1, 2, 3$ are 2, 4, 6 $\rightarrow$ right sum $= 1 \times (2 + 4 + 6) = 12$. True area: triangle with base 3, height $2 \times 3 = 6 \rightarrow \frac{1}{2} \times 3 \times 6 = 9$. Sandwich: $6 < 9 < 12$ ✓. **Answer: right sum = 12; true area = 9, which sits between 6 and 12.**

5. Interval $t = 2$ to $t = 6$, 8 equal slices: slice width and first three edges?

Work: interval length $= 6 - 2 = 4$ (NOT 6 — subtract the start!). Width $= 4 \div 8 = 0.5$. Edges start at 2 and step by 0.5. **Answer: width = 0.5; first three edges at $t = 2, 2.5, 3$.**

6. $y = x^2$ from $x = 0$ to 4, two slices of width 2: left and right sums.

Work: heights at edges 0, 2, 4 are 0, 4, 16. Left sum $= 2 \times (0 + 4) = 8$. Right sum $= 2 \times (4 + 16) = 40$. (Truth ≈ 21.3 — a very wide sandwich; only 2 fat slices.) **Answer: left = 8, right = 40.** (Watch for students forgetting to multiply by the width 2.)

7. Increasing speed graph: which sum is bigger and why?

Work: on an increasing graph, each slice's right edge is its highest point, so every right rectangle's top pokes ABOVE the curve (too much area), while every left rectangle's top sits BELOW the curve (too little). **Answer: the right sum is bigger — right rectangles overshoot a rising curve, left rectangles undershoot it.**

8. Draining tub, DECREASING rates 12, 9, 6, 4 L/min at $t = 0, 1, 2, 3$. Left and right sums for $t = 0$ to 3, and over/under labels.

Work: left sum $= 1 \times (12 + 9 + 6) = 27$; right sum $= 1 \times (9 + 6 + 4) = 19$. Decreasing graph flips the roles: left edges are the TALL ones $\rightarrow$ left = overestimate; right = underestimate. Truth between 19 and 27. **Answer: left = 27 (overestimate), right = 19 (underestimate).** (Classic error: assuming left is always the underestimate — that's only true for increasing graphs.)

9. Increasing graph, left = 30, right = 38; classmate claims the truth is 40.

Work: for an increasing graph the true total is guaranteed to be sandwiched: $30 \leq \text{truth} \leq 38$. Even the deliberately-too-big right sum only reached 38, so 40 is impossible. **Answer: 40 lies outside the sandwich [30, 38], so the classmate is wrong; the true total is some number between 30 and 38.**

10. $y = 2x$ from $x = 0$ to 4, two slices of width 2: left sum, right sum, their average, and the true area.

Work: heights at edges 0, 2, 4 are 0, 4, 8. Left $= 2 \times (0 + 4) = 8$. Right $= 2 \times (4 + 8) = 24$. Average $= (8 + 24) \div 2 = 16$. True area: triangle, $\frac{1}{2} \times 4 \times 8 = 16$. **Answer: left = 8, right = 24, average = 16 — exactly equal to the true area.** Averaging left and right is perfect for straight-line graphs (it builds trapezoids whose slanted tops lie right on the line); for curves it's merely very good.